BTPMA PLATFORM DOCUMENTATION

Education Ministry Folder Structure

educationministry.org — Complete Technical Reference

📁 educationministry.org/

Purpose: Educational Platform & Ministry Resources Website

Stack: Next.js 14 (App Router) • TypeScript • Tailwind CSS • MDX

Port: 3003 • **Document Version:** 1.0

Date: March 2026

Table of Contents

1. Project Overview & Purpose 3

2. High-Level Architecture 4

3. Complete Folder Structure 5

3.1 Root-Level Files & Configuration 5

3.2 Full Directory Tree 6

4. App Directory — Pages & Routing 8

4.1 Public Pages 8

4.2 Educational Content & Programs 9

4.3 Member Dashboard 10

5. API Routes 11

6. Components Library 12

7. Content Management 14

8. Technical Details & Configuration 15

9. Deployment & Infrastructure 17

10. Integration Points 19

11. Appendix — Quick Reference 20

1. Project Overview & Purpose

> **educationministry.org** is the flagship educational platform within the BTPMA ecosystem, providing ministry resources, course catalogs, community engagement tools, and the *Young Civic Architect Program* — a youth civic leadership initiative.

1.1 Mission & Goals

- **Educate** — Deliver structured courses, curricula, and learning paths for civic education
- **Empower** — Provide ministry resources that enable community leaders and educators
- **Engage** — Foster a community through events, blogs, newsletters, and member resources
- **Architect** — Run the Young Civic Architect Program for youth civic leadership

1.2 Target Audiences

Primary Audiences

- Students & youth participants
- Community educators & mentors
- Ministry leaders & administrators
- Parents & guardians

Secondary Audiences

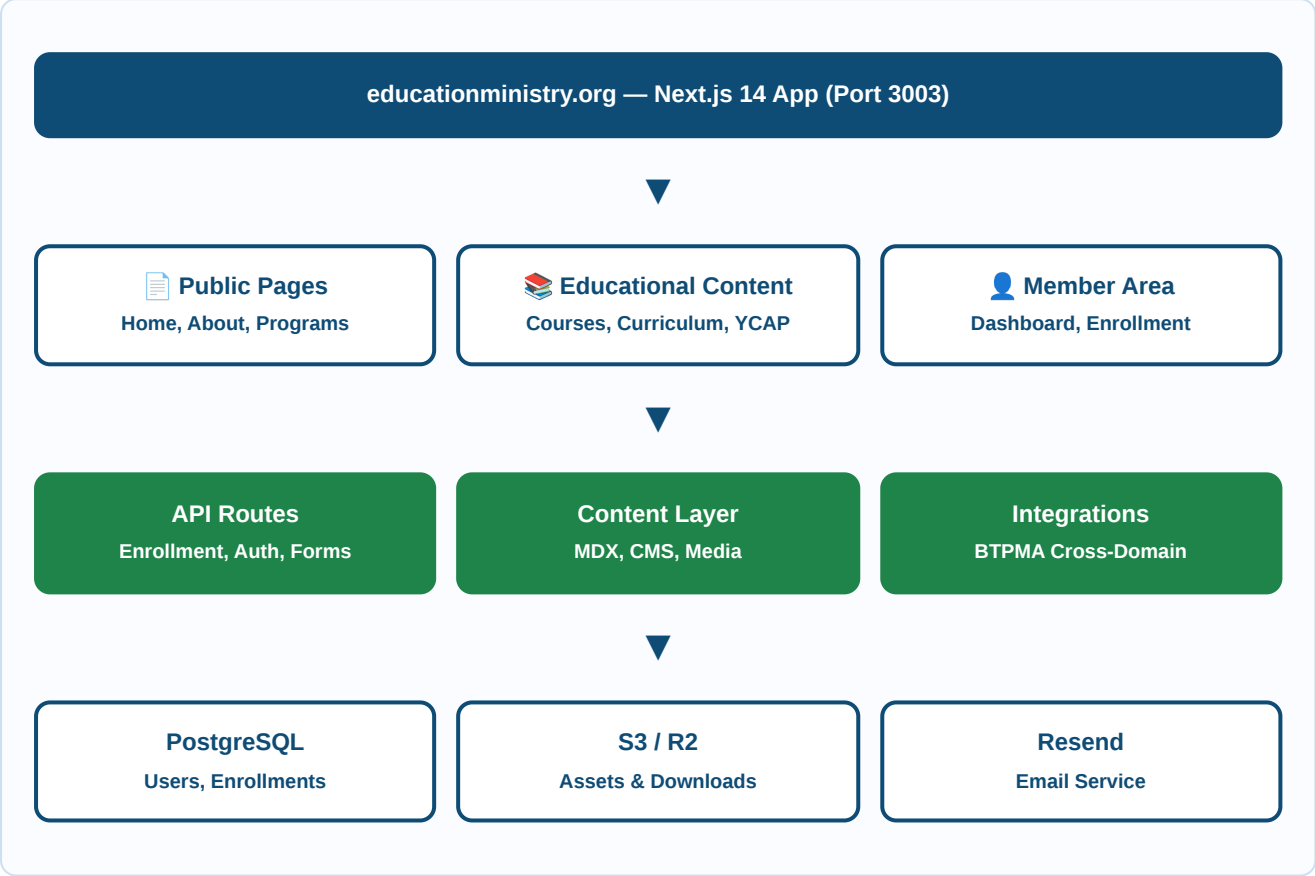
- Partner organizations
- Donors & sponsors
- General public seeking resources
- Media & press

1.3 Technology Stack

LAYER	TECHNOLOGY	PURPOSE
Framework	`Next.js 14` (App Router)	Full-stack React framework with SSR/SSG
Language	`TypeScript 5.x`	Type-safe development
Styling	`Tailwind CSS 3.x`	Utility-first responsive design
Content	`MDX` + Contentlayer	Markdown-based content with components
Database	`PostgreSQL` via Prisma ORM	User data, enrollments, events
Auth	`NextAuth.js`	Session-based authentication
Email	`Resend` + React Email	Transactional & newsletter emails
Storage	`AWS S3` / Cloudflare R2	Media assets & downloadable resources
Deployment	`Nginx` + PM2	Reverse proxy & process management

2. High-Level Architecture

2.1 Application Architecture Diagram



2.2 Request Flow

> **Client** → **Nginx (port 80/443)** → Reverse proxy to **Next.js (port 3003)** → App Router resolves page/API → Server Components render with data from **Prisma/PostgreSQL** & **MDX/Contentlayer** → HTML streamed to client.

2.3 Content Strategy

CONTENT TYPE	STORAGE	APPROACH
Static pages (About, Contact)	MDX files	Build-time rendering (SSG)
Course catalog & curriculum	MDX + DB metadata	ISR (Incremental Static Regeneration)
Blog articles & resources	MDX files	SSG with on-demand revalidation
Events & calendar	PostgreSQL	Server-side rendering (SSR)

User data & enrollments	PostgreSQL	SSR with authentication
Downloadable resources	S3 / R2	Signed URLs via API routes

3. Complete Folder Structure

3.1 Root-Level Files & Configuration

```
educationministry.org/
├── .env                        # Environment variables (DB, API keys, secrets)
├── .env.example                # Template with required env var documentation
├── .eslintrc.json              # ESLint configuration
├── .gitignore                  # Git ignore rules
├── .prettierrc                 # Prettier code formatting config
├── contentlayer.config.ts      # Contentlayer MDX processing config
├── middleware.ts               # Next.js middleware (auth, redirects, i18n)
├── next.config.js              # Next.js configuration
├── package.json                # Dependencies and scripts
├── postcss.config.js           # PostCSS / Tailwind processing
├── prisma/
│   ├── schema.prisma           # Database schema definition
│   ├── seed.ts                 # Database seed data
│   └── migrations/             # Database migration history
├── tailwind.config.ts          # Tailwind CSS configuration
├── tsconfig.json               # TypeScript configuration
└── README.md                   # Project documentation
```

Key Configuration Files Explained

FILE	PURPOSE
`middleware.ts`	Intercepts requests for auth checks, redirects unauthenticated users from `/dashboard/*`, handles domain-aware routing
`contentlayer.config.ts`	Defines content schemas for courses, articles, resources; processes MDX at build time
`prisma/schema.prisma`	Defines User, Course, Enrollment, Event, Resource, Newsletter models
`next.config.js`	Image domains, MDX plugins, redirects, headers, environment bindings

3.2 Full Directory Tree

```
educationministry.org/
│
├── public/                          # Static assets served at root
│   ├── favicon.ico
│   ├── robots.txt
│   ├── sitemap.xml
│   ├── manifest.json
│   ├── images/
│   │   ├── logo.svg
│   │   ├── logo-dark.svg
│   │   ├── og-image.jpg               # Open Graph default image
│   │   ├── hero/                      # Hero/banner images
│   │   │   ├── home-hero.webp
│   │   │   ├── programs-hero.webp
│   │   │   └── ycap-hero.webp
│   │   ├── programs/                  # Program imagery
│   │   ├── team/                      # Staff & instructor photos
│   │   ├── events/                    # Event photos
│   │   └── icons/                     # UI icons & illustrations
│   ├── fonts/
│   │   ├── inter-var.woff2
│   │   └── playfair-display.woff2
│   ├── documents/                     # Publicly downloadable docs
│   │   ├── annual-report-2026.pdf
│   │   ├── program-brochure.pdf
│   │   └── enrollment-guide.pdf
│   └── videos/
│       └── intro-welcome.mp4
│
├── src/
│   ├── app/                           # Next.js App Router
│   │   ├── layout.tsx                 # Root layout (html, body, providers)
│   │   ├── page.tsx                   # Homepage
│   │   ├── loading.tsx                # Global loading state
│   │   ├── error.tsx                  # Global error boundary
│   │   ├── not-found.tsx              # Custom 404 page
│   │   ├── globals.css                # Global styles + Tailwind directives
│   │   │
│   │   ├── (public)/                  # Public route group
│   │   │   ├── about/
│   │   │   │   ├── page.tsx           # About the ministry
│   │   │   │   ├── mission/page.tsx   # Mission & vision
│   │   │   │   ├── team/page.tsx      # Leadership & staff
│   │   │   │   └── history/page.tsx   # Organization history
│   │   │   ├── contact/
│   │   │   │   └── page.tsx           # Contact form & info
│   │   │   ├── faq/
│   │   │   │   └── page.tsx           # Frequently asked questions
│   │   │   ├── donate/
│   │   │   │   └── page.tsx           # Donation/support page
```

```
│   │   │   └── privacy/
│   │   │       └── page.tsx            # Privacy policy
```

```
│   │   │
│   │   ├── programs/                    # Educational programs
│   │   │   ├── page.tsx                 # Programs overview/catalog
│   │   │   ├── layout.tsx               # Programs section layout
│   │   │   ├── [slug]/
│   │   │   │   └── page.tsx             # Individual program detail
│   │   │   ├── civic-education/
│   │   │   │   └── page.tsx             # Civic education track
│   │   │   ├── leadership-development/
│   │   │   │   └── page.tsx             # Leadership program
│   │   │   └── community-organizing/
│   │   │       └── page.tsx             # Community organizing track
│   │   │
│   │   ├── courses/                     # Course catalog
│   │   │   ├── page.tsx                 # All courses listing
│   │   │   ├── [courseId]/
│   │   │   │   ├── page.tsx             # Course detail/overview
│   │   │   │   ├── curriculum/
│   │   │   │   │   └── page.tsx         # Full curriculum view
│   │   │   │   ├── enroll/
│   │   │   │   │   └── page.tsx         # Enrollment form
│   │   │   │   └── [lessonId]/
│   │   │   │       └── page.tsx         # Individual lesson view
│   │   │   └── categories/
│   │   │       ├── page.tsx             # Course categories
│   │   │       └── [category]/
│   │   │           └── page.tsx         # Filtered by category
│   │   │
│   │   ├── ycap/                        # Young Civic Architect Program
│   │   │   ├── page.tsx                 # YCAP landing page
│   │   │   ├── about/page.tsx           # About YCAP
│   │   │   ├── apply/page.tsx           # Application form
│   │   │   ├── curriculum/page.tsx      # YCAP curriculum overview
│   │   │   ├── mentors/page.tsx         # Mentor directory
│   │   │   ├── projects/
│   │   │   │   ├── page.tsx             # Student projects showcase
│   │   │   │   └── [projectId]/
│   │   │   │       └── page.tsx         # Individual project
│   │   │   └── alumni/page.tsx          # Alumni stories
│   │   │
│   │   ├── resources/                   # Resource library
│   │   │   ├── page.tsx                 # Resource hub/index
│   │   │   ├── layout.tsx               # Resources layout with sidebar
│   │   │   ├── [resourceId]/
│   │   │   │   └── page.tsx             # Individual resource view
│   │   │   ├── guides/page.tsx          # How-to guides
│   │   │   ├── toolkits/page.tsx        # Downloadable toolkits
│   │   │   ├── videos/page.tsx          # Video library
│   │   │   ├── worksheets/page.tsx      # Educational worksheets
│   │   │   └── categories/
```

```
│   │   │       └── [category]/page.tsx  # Filter by category
```

```
│   │   │
│   │   ├── blog/                       # Blog & articles
│   │   │   ├── page.tsx                # Blog listing (paginated)
│   │   │   ├── [slug]/page.tsx         # Individual article
│   │   │   ├── categories/
│   │   │   │   ├── page.tsx            # All categories
│   │   │   │   └── [category]/page.tsx  # Posts by category
│   │   │   └── authors/
│   │   │       ├── page.tsx            # All authors
│   │   │       └── [authorId]/page.tsx  # Author profile + posts
│   │   │
│   │   ├── events/                     # Events & calendar
│   │   │   ├── page.tsx                # Upcoming events list
│   │   │   ├── [eventId]/page.tsx      # Event detail + registration
│   │   │   ├── calendar/page.tsx       # Calendar view
│   │   │   └── past/page.tsx           # Past events archive
│   │   │
│   │   ├── community/                  # Community section
│   │   │   ├── page.tsx                # Community hub
│   │   │   ├── members/page.tsx        # Member directory
│   │   │   ├── forums/page.tsx         # Discussion forums
│   │   │   └── success-stories/
│   │   │       ├── page.tsx            # Success stories listing
│   │   │       └── [storyId]/page.tsx   # Individual story
│   │   │
│   │   ├── dashboard/                  # Authenticated member area
│   │   │   ├── layout.tsx              # Dashboard layout (sidebar nav)
│   │   │   ├── page.tsx                # Dashboard home/overview
│   │   │   ├── courses/
│   │   │   │   ├── page.tsx            # My enrolled courses
│   │   │   │   └── [courseId]/
│   │   │   │       ├── page.tsx        # Course progress view
│   │   │   │       └── [lessonId]/
│   │   │   │           └── page.tsx    # Lesson player/viewer
│   │   │   ├── profile/page.tsx        # User profile settings
│   │   │   ├── certificates/page.tsx   # Earned certificates
│   │   │   ├── bookmarks/page.tsx      # Saved resources
│   │   │   └── notifications/page.tsx  # User notifications
│   │   │
│   │   ├── auth/                       # Authentication pages
│   │   │   ├── login/page.tsx          # Login page
│   │   │   ├── register/page.tsx       # Registration page
│   │   │   ├── forgot-password/page.tsx # Password recovery
│   │   │   └── verify/page.tsx         # Email verification
│   │   │
│   │   └── api/                        # API routes (Section 5)
│   │       └── ...
```

```
│   │
│   ├── components/                    # Reusable components (Section 6)
│   │   ├── ui/                        # Base UI primitives
│   │   ├── layout/                    # Layout components
│   │   ├── education/                 # Educational components
│   │   ├── resources/                 # Resource components
│   │   ├── blog/                      # Blog components
│   │   ├── events/                    # Event components
│   │   ├── community/                 # Community components
│   │   ├── forms/                     # Form components
│   │   ├── dashboard/                 # Dashboard components
│   │   └── shared/                    # Cross-cutting components
│   │
│   ├── lib/                           # Shared utilities & config
│   │   ├── prisma.ts                  # Prisma client singleton
│   │   ├── auth.ts                    # NextAuth configuration
│   │   ├── utils.ts                   # General utility functions
│   │   ├── constants.ts               # App-wide constants
│   │   ├── validations.ts            # Zod schemas for form validation
│   │   ├── email.ts                   # Email sending utilities
│   │   ├── storage.ts                 # S3/R2 file operations
│   │   └── seo.ts                     # SEO metadata helpers
│   │
│   ├── hooks/                         # Custom React hooks
│   │   ├── use-auth.ts                # Authentication state hook
│   │   ├── use-enrollment.ts          # Course enrollment logic
│   │   ├── use-search.ts             # Search functionality
│   │   ├── use-pagination.ts          # Pagination state
│   │   ├── use-media-query.ts         # Responsive breakpoints
│   │   └── use-local-storage.ts       # Persistent client state
│   │
│   ├── types/                         # TypeScript type definitions
│   │   ├── index.ts                   # Barrel export
│   │   ├── course.ts                  # Course & lesson types
│   │   ├── user.ts                    # User & profile types
│   │   ├── event.ts                   # Event types
│   │   ├── resource.ts                # Resource types
│   │   ├── blog.ts                    # Blog & article types
│   │   └── api.ts                     # API request/response types
│   │
│   ├── styles/                        # Additional styles
│   │   ├── mdx.css                    # MDX content typography
│   │   ├── calendar.css               # Calendar component styles
│   │   └── animations.css             # Shared animations
│   │
│   └── config/                        # App configuration
│       ├── site.ts                    # Site metadata (name, desc, URLs)
│       ├── navigation.ts              # Nav menu structure
│       └── programs.ts                # Program definitions & metadata
```

```
│
├── content/                            # MDX content files
│   ├── courses/                        # Course content (MDX)
│   │   ├── civic-foundations/
│   │   │   ├── index.mdx               # Course overview
│   │   │   ├── lesson-01-intro.mdx     # Lesson content
│   │   │   ├── lesson-02-government.mdx
│   │   │   └── ...
│   │   ├── leadership-101/
│   │   │   ├── index.mdx
│   │   │   └── ...
│   │   └── community-action/
│   │       ├── index.mdx
│   │       └── ...
│   │
│   ├── blog/                           # Blog posts (MDX)
│   │   ├── 2026-03-civic-engagement.mdx
│   │   ├── 2026-02-youth-leadership.mdx
│   │   └── ...
│   │
│   ├── resources/                      # Resource descriptions (MDX)
│   │   ├── guides/
│   │   │   ├── community-organizing-guide.mdx
│   │   │   └── voter-education-toolkit.mdx
│   │   ├── toolkits/
│   │   │   └── civic-action-toolkit.mdx
│   │   └── worksheets/
│   │       └── civic-engagement-worksheet.mdx
│   │
│   ├── pages/                          # Static page content (MDX)
│   │   ├── about.mdx
│   │   ├── mission.mdx
│   │   ├── faq.mdx
│   │   └── privacy.mdx
│   │
│   └── ycap/                           # YCAP program content
│       ├── overview.mdx
│       ├── curriculum.mdx
│       ├── mentor-bios/
│       │   ├── mentor-jane-doe.mdx
│       │   └── ...
│       └── projects/
│           ├── project-community-garden.mdx
│           └── ...
│
├── emails/                             # React Email templates
│   ├── welcome.tsx                     # Welcome email
│   ├── enrollment-confirmation.tsx     # Course enrollment confirm
│   ├── event-reminder.tsx              # Event reminder
│   ├── newsletter.tsx                  # Newsletter template
│   ├── password-reset.tsx              # Password reset email
│   └── certificate.tsx                 # Certificate issuance
│
```

```
└── scripts/                          # Utility scripts
    ├── seed-courses.ts               # Seed course data
    ├── generate-sitemap.ts           # Dynamic sitemap generation
    ├── sync-content.ts               # Content sync utilities
    └── backup-db.sh                  # Database backup script
```

4. App Directory — Pages & Routing

4.1 Public Pages

Public pages are accessible without authentication and serve as the main marketing and informational surface of the platform.

ROUTE	FILE	DESCRIPTION
`/`	`app/page.tsx`	Homepage — hero, featured programs, testimonials, CTA
`/about`	`app/(public)/about/page.tsx`	Organization overview, mission introduction
`/about/mission`	`app/(public)/about/mission/page.tsx`	Detailed mission, vision, and values
`/about/team`	`app/(public)/about/team/page.tsx`	Leadership team, instructors, staff bios
`/about/history`	`app/(public)/about/history/page.tsx`	Organization timeline and milestones
`/contact`	`app/(public)/contact/page.tsx`	Contact form, office locations, social links
`/faq`	`app/(public)/faq/page.tsx`	Accordion-style FAQ page
`/donate`	`app/(public)/donate/page.tsx`	Donation page with payment integration
`/privacy`	`app/(public)/privacy/page.tsx`	Privacy policy (rendered from MDX)

4.2 Educational Content & Programs

Programs Section

ROUTE	DESCRIPTION	RENDERING
`/programs`	All programs overview with filtering	SSG + ISR
`/programs/[slug]`	Dynamic program detail page	SSG (generateStaticParams)
`/programs/civic-education`	Civic education track landing page	SSG
`/programs/leadership-development`	Leadership program landing page	SSG
`/programs/community-organizing`	Community organizing track	SSG

Course Catalog

ROUTE	DESCRIPTION	RENDERING
`/courses`	Full course catalog with search & filters	SSR

`/courses/[courseId]`	Course overview, description, instructor	ISR (60s)
`/courses/[courseId]/curriculum`	Detailed curriculum outline	ISR
`/courses/[courseId]/enroll`	Enrollment form (may require auth)	SSR
`/courses/[courseId]/[lessonId]`	Lesson content viewer (auth required)	SSR
`/courses/categories`	Browse by category	SSG
`/courses/categories/[category]`	Filtered course listing	ISR

Young Civic Architect Program (YCAP)

> **YCAP** is the flagship youth initiative — a dedicated program section with its own application flow, mentor directory, curriculum, and student project showcase.

ROUTE	DESCRIPTION	RENDERING
`/ycap`	YCAP landing page with program overview	SSG
`/ycap/about`	Detailed program description & eligibility	SSG
`/ycap/apply`	Multi-step application form	SSR
`/ycap/curriculum`	YCAP curriculum modules & timeline	SSG
`/ycap/mentors`	Mentor directory with bios & expertise	ISR
`/ycap/projects`	Showcase of student civic projects	ISR
`/ycap/projects/[projectId]`	Individual project detail	ISR
`/ycap/alumni`	Alumni stories & outcomes	SSG

4.3 Member Dashboard

Protected routes requiring authentication. The dashboard layout includes a persistent sidebar navigation.

ROUTE	DESCRIPTION	FEATURES
`/dashboard`	Dashboard home	Progress overview, upcoming events, recommendations
`/dashboard/courses`	My courses	Enrolled courses with progress bars
`/dashboard/courses/[id]`	Course progress	Lesson list with completion status
`/dashboard/courses/[id]/[lessonId]`	Lesson player	Content viewer, video player, quiz
`/dashboard/profile`	Profile settings	Edit name, email, password, avatar
`/dashboard/certificates`	My certificates	View & download earned certificates
`/dashboard/bookmarks`	Saved items	Bookmarked resources & courses
`/dashboard/notifications`	Notifications	System alerts, course updates, events

Authentication Pages

ROUTE	DESCRIPTION

`/auth/login`	Email/password login with social OAuth options
`/auth/register`	New user registration with role selection
`/auth/forgot-password`	Password reset request (sends email)
`/auth/verify`	Email verification callback

5. API Routes

> All API routes live under `src/app/api/` and follow RESTful conventions. They use Next.js Route Handlers with proper HTTP method handling, input validation (Zod), and appropriate error responses.

```
src/app/api/
├── auth/
│   └── [...nextauth]/
│       └── route.ts              # NextAuth.js handler (GET, POST)
│
├── courses/
│   ├── route.ts                  # GET: list courses (with filters, pagination)
│   └── [courseId]/
│       ├── route.ts              # GET: course detail
│       ├── enroll/
│       │   └── route.ts          # POST: enroll in course
│       └── progress/
│           └── route.ts          # GET/PATCH: track lesson progress
│
├── resources/
│   ├── route.ts                  # GET: list resources (search, filter)
│   └── [resourceId]/
│       ├── route.ts              # GET: resource detail
│       └── download/
│           └── route.ts          # GET: generate signed download URL
│
├── contact/
│   └── route.ts                  # POST: contact form submission
│
├── newsletter/
│   ├── subscribe/
│   │   └── route.ts              # POST: subscribe to newsletter
│   └── unsubscribe/
│       └── route.ts              # POST: unsubscribe
│
├── events/
│   ├── route.ts                  # GET: list upcoming events
│   └── [eventId]/
│       ├── route.ts              # GET: event detail
│       └── register/
│           └── route.ts          # POST: register for event
│
├── ycap/
│   └── apply/
│       └── route.ts              # POST: YCAP application submission
│
├── users/
│   └── [userId]/
│       ├── route.ts              # GET/PATCH: user profile
│       ├── bookmarks/
```

```
│       │   └── route.ts          # GET/POST/DELETE: manage bookmarks
│       └── certificates/
│           └── route.ts          # GET: list certificates
│
├── search/
│   └── route.ts                  # GET: global site search
│
└── webhooks/
    ├── payment/
    │   └── route.ts              # POST: payment provider webhook
    └── email/
        └── route.ts              # POST: email delivery webhook
```

API Route Summary

ENDPOINT	METHOD	AUTH	DESCRIPTION
`/api/courses`	GET	Public	List courses with pagination & filters
`/api/courses/[id]/enroll`	POST	Required	Enroll authenticated user in a course
`/api/courses/[id]/progress`	PATCH	Required	Update lesson completion progress
`/api/resources/[id]/download`	GET	Optional	Generate signed URL for resource download
`/api/contact`	POST	Public	Submit contact form (rate-limited)
`/api/newsletter/subscribe`	POST	Public	Add email to newsletter list
`/api/events/[id]/register`	POST	Required	Register for an event
`/api/ycap/apply`	POST	Optional	Submit YCAP application
`/api/search`	GET	Public	Full-text search across site content

6. Components Library

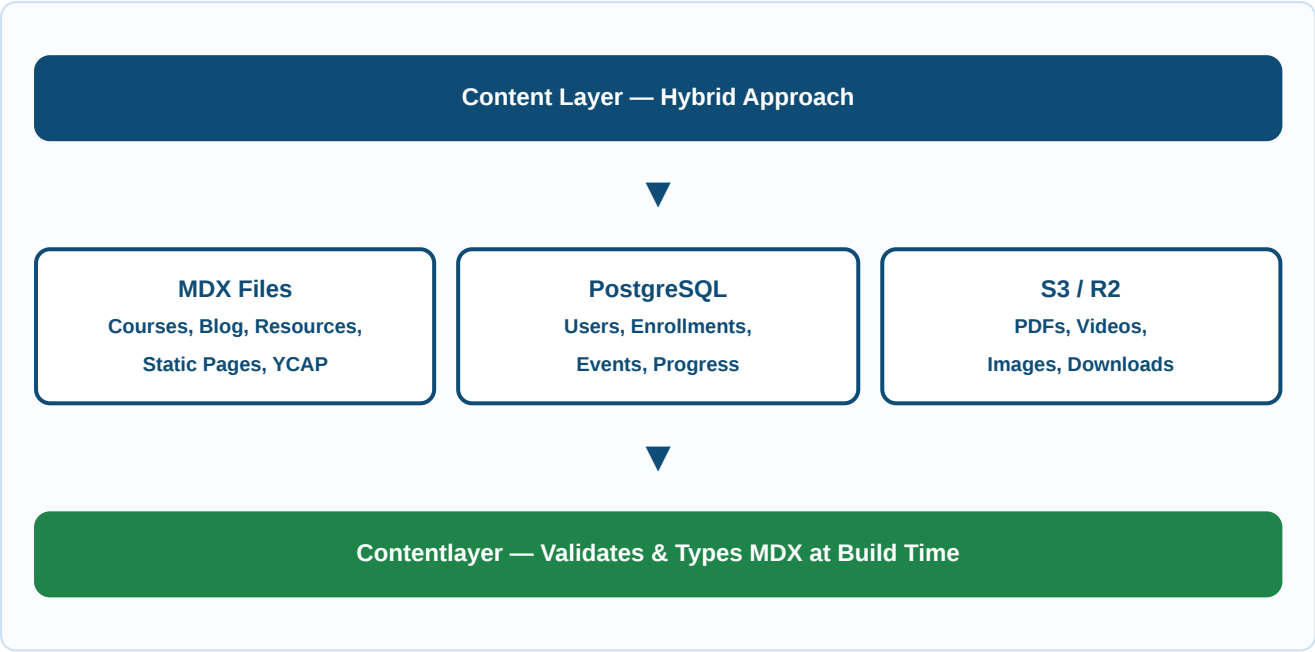
```
src/components/
│
├── ui/                             # Base UI primitives (design system)
│   ├── button.tsx                  # Button variants (primary, outline, ghost)
│   ├── card.tsx                    # Content card wrapper
│   ├── badge.tsx                   # Status/category badges
│   ├── input.tsx                   # Text input with label & error
│   ├── textarea.tsx                # Multi-line text input
│   ├── select.tsx                  # Dropdown select
│   ├── modal.tsx                   # Modal/dialog overlay
│   ├── tabs.tsx                    # Tab navigation
│   ├── accordion.tsx               # Collapsible accordion
│   ├── progress-bar.tsx            # Progress indicator
│   ├── avatar.tsx                  # User avatar (image/initials)
│   ├── skeleton.tsx                # Loading skeleton placeholders
│   ├── toast.tsx                   # Notification toasts
│   ├── pagination.tsx              # Page navigation
│   └── breadcrumb.tsx              # Breadcrumb navigation
│
├── layout/                         # Structural layout components
│   ├── header.tsx                  # Site header with nav & mobile menu
│   ├── footer.tsx                  # Site footer with links & newsletter
│   ├── sidebar.tsx                 # Sidebar navigation (resources, dashboard)
│   ├── mobile-nav.tsx              # Mobile slide-out navigation
│   ├── container.tsx               # Max-width page container
│   ├── section.tsx                 # Content section with title
│   └── page-header.tsx             # Page title, breadcrumb, description
│
├── education/                      # Educational/course components
│   ├── course-card.tsx             # Course preview card
│   ├── course-grid.tsx             # Grid layout for course cards
│   ├── course-detail-hero.tsx      # Course detail header
│   ├── curriculum-display.tsx      # Curriculum outline with modules
│   ├── lesson-viewer.tsx           # Lesson content renderer (MDX)
│   ├── lesson-nav.tsx              # Previous/next lesson navigation
│   ├── progress-tracker.tsx        # Course progress sidebar
│   ├── quiz-component.tsx          # Interactive quiz/assessment
│   ├── video-player.tsx            # Embedded video player
│   ├── instructor-card.tsx         # Instructor bio card
│   └── certificate-badge.tsx       # Course completion certificate
```

```
│
├── resources/                      # Resource library components
│   ├── resource-card.tsx           # Resource preview card
│   ├── resource-grid.tsx           # Filterable resource grid
│   ├── resource-detail.tsx         # Full resource view
│   ├── document-viewer.tsx         # Inline document/PDF viewer
│   ├── download-button.tsx         # Download with tracking
│   ├── resource-filter.tsx         # Category/type filter bar
│   └── resource-search.tsx         # Resource search input
│
├── blog/                           # Blog & article components
│   ├── article-card.tsx            # Blog post preview card
│   ├── article-grid.tsx            # Blog listing grid/list
│   ├── article-header.tsx          # Article title, author, date
│   ├── article-body.tsx            # MDX content renderer
│   ├── author-bio.tsx              # Author info box
│   ├── category-chips.tsx          # Category tag pills
│   ├── share-buttons.tsx           # Social sharing buttons
│   ├── related-posts.tsx           # Related articles section
│   └── table-of-contents.tsx       # Auto-generated TOC sidebar
│
├── events/                         # Event components
│   ├── event-card.tsx              # Event preview card
│   ├── event-list.tsx              # Upcoming events list
│   ├── event-detail.tsx            # Full event detail view
│   ├── event-calendar.tsx          # Monthly calendar view
│   ├── event-registration.tsx      # Registration form
│   ├── event-countdown.tsx         # Countdown timer
│   └── event-map.tsx               # Location map embed
│
├── community/                      # Community components
│   ├── member-card.tsx             # Member profile card
│   ├── success-story.tsx           # Testimonial/story card
│   ├── community-stats.tsx         # Impact statistics display
│   └── discussion-thread.tsx       # Forum thread component
│
├── forms/                          # Form components
│   ├── contact-form.tsx            # Contact page form
│   ├── enrollment-form.tsx         # Course enrollment form
│   ├── ycap-application.tsx        # Multi-step YCAP application
│   ├── event-registration-form.tsx # Event RSVP form
│   ├── newsletter-signup.tsx       # Email newsletter form
│   ├── search-form.tsx             # Global search input
│   └── donation-form.tsx           # Donation/payment form
│
├── dashboard/                      # Dashboard-specific components
│   ├── dashboard-nav.tsx           # Dashboard sidebar navigation
│   ├── stats-cards.tsx             # Overview statistics cards
│   ├── enrolled-course-card.tsx    # Course card with progress
│   ├── notification-list.tsx       # Notification feed
│   └── activity-feed.tsx           # Recent activity timeline
│
└── shared/                         # Cross-cutting components
```

```
    ├── mdx-components.tsx          # Custom MDX rendering components
    ├── seo-head.tsx                # Dynamic SEO meta tags
    ├── theme-provider.tsx          # Light/dark theme provider
    ├── analytics.tsx               # Analytics tracking wrapper
    ├── error-boundary.tsx          # Error boundary wrapper
    ├── back-to-top.tsx             # Scroll to top button
    └── loading-spinner.tsx         # Reusable loading spinner
```

7. Content Management

7.1 Content Architecture



7.2 MDX Frontmatter Schemas

Course Schema

```
---
title: "Civic Foundations 101"
slug: "civic-foundations-101"
description: "An introductory course on civic principles and engagement."
category: "civic-education"
level: "beginner"              # beginner | intermediate | advanced
duration: "6 weeks"
instructor: "Dr. Jane Smith"
thumbnail: "/images/programs/civic-101.webp"
published: true
featured: true
prerequisites: []
tags: ["civics", "government", "citizenship"]
createdAt: "2026-01-15"
updatedAt: "2026-03-10"
---
```

Blog Post Schema

```
---
title: "Empowering Youth Through Civic Education"
```

```
slug: "empowering-youth-civic-education"
author: "Dr. Jane Smith"
authorId: "jane-smith"
category: "civic-engagement"
description: "How structured civic education transforms youth participation."
thumbnail: "/images/blog/youth-civic.webp"
published: true
featured: false
tags: ["youth", "education", "civic-engagement"]
publishedAt: "2026-03-15"
---
```

7.3 Media Assets Organization

ASSET TYPE	LOCATION	FORMAT	NOTES
Hero images	`public/images/hero/`	WebP, 1920×800	Optimized via Next/Image
Thumbnails	`public/images/programs/`	WebP, 800×450	Course & program cards
Team photos	`public/images/team/`	WebP, 400×400	Square aspect for avatars
Documents	`S3 bucket`	PDF	Served via signed URLs
Videos	`S3 bucket` or CDN	MP4, HLS	Streamed via video player
Icons & illustrations	`public/images/icons/`	SVG	Inline or component-based

8. Technical Details & Configuration

8.1 Next.js App Router Structure

> The project uses Next.js 14 App Router with **React Server Components** by default. Client components are explicitly marked with `"use client"`. Route groups `(public)` organize pages without affecting URL paths.

Key Architectural Decisions

- **Server Components First** — All pages render on server by default for SEO and performance
- **Streaming SSR** — Loading states with `loading.tsx` enable instant page transitions
- **Route Groups** — `(public)` groups public pages sharing a layout; `dashboard` has its own auth-protected layout
- **Parallel Routes** — Dashboard uses parallel routes for modals and slide-overs
- **ISR** — Course pages use Incremental Static Regeneration (60s revalidation) for performance + freshness

8.2 Authentication Strategy

```
// src/lib/auth.ts — NextAuth.js Configuration
import NextAuth from "next-auth";
import CredentialsProvider from "next-auth/providers/credentials";
import GoogleProvider from "next-auth/providers/google";
import { PrismaAdapter } from "@auth/prisma-adapter";
import { prisma } from "@/lib/prisma";

export const authOptions = {
  adapter: PrismaAdapter(prisma),
  providers: [
    CredentialsProvider({ /* email/password */ }),
    GoogleProvider({ clientId: ..., clientSecret: ... }),
  ],
  session: { strategy: "jwt" },
  pages: {
    signIn: "/auth/login",
    newUser: "/auth/register",
    error: "/auth/error",
  },
  callbacks: {
    session: ({ session, token }) => ({
      ...session,
      user: { ...session.user, id: token.sub, role: token.role },
    }),
  },
```

```
};
```

8.3 SEO Optimization

FEATURE	IMPLEMENTATION
Dynamic Metadata	`generateMetadata()` per page with title, description, OG tags
Sitemap	`app/sitemap.ts` — dynamic sitemap from MDX + DB content
Robots	`public/robots.txt` — allow all, reference sitemap
Structured Data	JSON-LD for courses (`Course`), events (`Event`), articles (`Article`)
Canonical URLs	Auto-generated canonical tags for all pages
Open Graph	Per-page OG images, Twitter cards, description

8.4 Performance Optimization

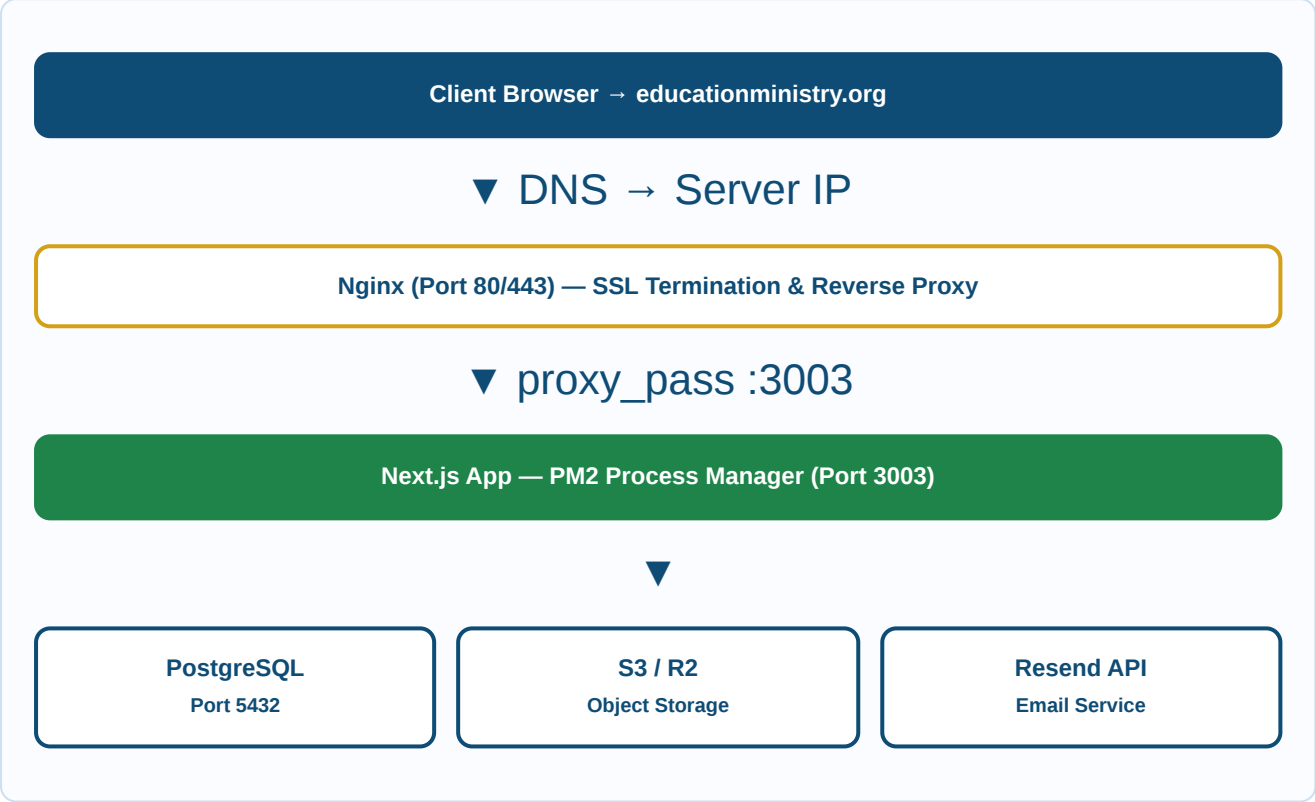
- **Image Optimization** — `next/image` with WebP, responsive sizes, lazy loading, blur placeholders
- **Font Optimization** — Self-hosted variable fonts via `next/font`, no layout shift
- **Code Splitting** — Automatic per-route splitting; dynamic imports for heavy components (video player, calendar)
- **Edge Caching** — Static pages cached at CDN edge; ISR pages cached with stale-while-revalidate
- **Bundle Analysis** — `@next/bundle-analyzer` integrated for tree-shaking verification
- **Database** — Prisma query optimization, connection pooling, indexed queries

8.5 Responsive Design

BREAKPOINT	TAILWIND CLASS	TARGET
0–639px	`default`	Mobile phones (portrait)
640–767px	`sm:`	Large phones (landscape)
768–1023px	`md:`	Tablets
1024–1279px	`lg:`	Small laptops
1280px+	`xl:`	Desktops & large screens

9. Deployment & Infrastructure

9.1 Deployment Architecture



9.2 Nginx Configuration

```
# /etc/nginx/sites-available/educationministry.org

server {
    listen 80;
    server_name educationministry.org www.educationministry.org;
    return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name educationministry.org www.educationministry.org;

    ssl_certificate     /etc/letsencrypt/live/educationministry.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/educationministry.org/privkey.pem;
    ssl_protocols       TLSv1.2 TLSv1.3;
    ssl_ciphers         HIGH:!aNULL:!MD5;

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;
```

```
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains" always;
    add_header Referrer-Policy "strict-origin-when-cross-origin" always;

    # Gzip compression
    gzip on;
    gzip_types text/plain text/css application/json application/javascript
               text/xml application/xml image/svg+xml;
    gzip_min_length 1000;

    # Static assets with long cache
    location /_next/static/ {
        proxy_pass http://127.0.0.1:3003;
        expires 365d;
        add_header Cache-Control "public, immutable";
    }

    location /images/ {
        alias /var/www/educationministry.org/public/images/;
        expires 30d;
        add_header Cache-Control "public";
    }

    # Main proxy
    location / {
        proxy_pass         http://127.0.0.1:3003;
        proxy_http_version 1.1;
        proxy_set_header   Upgrade $http_upgrade;
        proxy_set_header   Connection "upgrade";
        proxy_set_header   Host $host;
        proxy_set_header   X-Real-IP $remote_addr;
        proxy_set_header   X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header   X-Forwarded-Proto $scheme;
        proxy_buffer_size  128k;
        proxy_buffers      4 256k;
    }
}
```

9.3 PM2 Process Configuration

```
// ecosystem.config.js
module.exports = {
  apps: [{
    name: "educationministry",
    script: "node_modules/.bin/next",
    args: "start -p 3003",
    cwd: "/var/www/educationministry.org",
    instances: 2,                    // 2 instances for load balancing
    exec_mode: "cluster",
    env: {
      NODE_ENV: "production",
      PORT: 3003,
      DATABASE_URL: "postgresql://...",
      NEXTAUTH_URL: "https://educationministry.org",
      NEXTAUTH_SECRET: "...",
    },
    max_memory_restart: "500M",
    log_date_format: "YYYY-MM-DD HH:mm:ss",
    error_file: "/var/log/pm2/educationministry-error.log",
    out_file: "/var/log/pm2/educationministry-out.log",
  }],
};
```

9.4 SSL / HTTPS Setup

```
# Install Certbot and obtain SSL certificate
sudo apt install certbot python3-certbot-nginx
sudo certbot --nginx -d educationministry.org -d www.educationministry.org

# Auto-renewal cron (certbot auto-configures, verify with):
sudo certbot renew --dry-run

# Renewal cron entry (auto-created):
# 0 0,12 * * * certbot renew --quiet --post-hook "systemctl reload nginx"
```

9.5 Port Configuration Summary

SERVICE	PORT	ACCESS	NOTES
Nginx HTTP	80	Public	Redirects to HTTPS
Nginx HTTPS	443	Public	SSL termination
Next.js App	3003	Internal	Proxied via Nginx; **educationministry.org**

PostgreSQL	5432	Internal	Localhost only; shared or dedicated instance

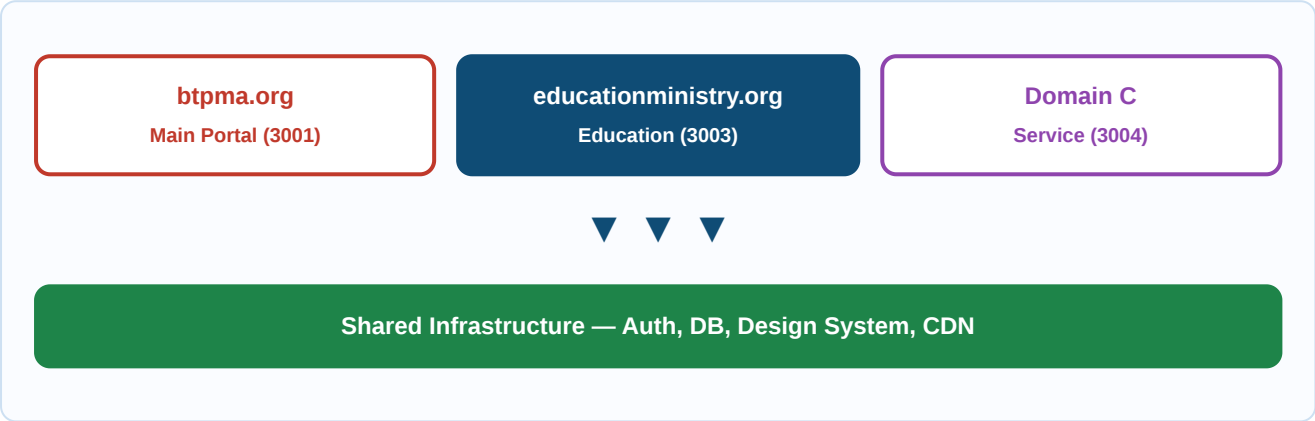
9.6 Build & Deploy Commands

```
# Build
cd /var/www/educationministry.org
npm ci --production=false
npx prisma generate
npx prisma migrate deploy
npm run build                        # Builds Next.js + Contentlayer

# Deploy
pm2 start ecosystem.config.js
pm2 save
sudo systemctl reload nginx

# Zero-downtime update
git pull origin main
npm ci && npm run build
pm2 reload educationministry
```

10. Integration Points

10.1 BTPMA Ecosystem Overview

educationministry.org operates as part of the broader BTPMA multi-domain platform. Each domain runs independently but shares authentication, navigation, and design language.



10.2 Cross-Domain Integration

INTEGRATION	MECHANISM	DETAILS
Shared Auth	JWT + shared secret	Users authenticate once; JWT tokens valid across all BTPMA domains via shared `NEXTAUTH_SECRET`
Cross-domain Nav	Shared header config	Common navigation component pulls menu structure from `@btpma/shared-config` package
Shared Design	NPM package	`@btpma/ui` shared component library ensures visual consistency
Shared DB	PostgreSQL schemas	Shared `users` table; domain-specific schemas for courses, events, etc.
API Gateway	Nginx routing	Each domain listens on its own port; Nginx routes by `server_name`
Analytics	Shared tracker	Common analytics script reports to centralized dashboard

10.3 Shared Packages

```
# Monorepo shared packages (or published NPM packages)
packages/
├── @btpma/ui/                  # Shared UI components
│   ├── header.tsx              # Universal header with domain-aware nav
│   ├── footer.tsx              # Universal footer
│   ├── theme.ts                # Design tokens (colors, fonts, spacing)
│   └── ...
```

```
├── @btpma/shared-config/        # Shared configuration
│   ├── navigation.ts            # Cross-domain menu structure
│   ├── domains.ts               # Domain registry (URLs, ports, names)
│   └── seo-defaults.ts          # Default SEO configuration
└── @btpma/auth/                 # Shared authentication
    ├── session.ts               # Session validation helpers
    ├── middleware.ts             # Auth middleware for cross-domain
    └── types.ts                 # Shared user/session types
```

10.4 Domain-Aware Routing

```
// middleware.ts — Domain routing excerpt
import { NextResponse } from "next/server";
import type { NextRequest } from "next/server";

export function middleware(request: NextRequest) {
  const host = request.headers.get("host") || "";

  // Ensure this app only handles educationministry.org
  if (!host.includes("educationministry.org") && !host.includes("localhost:3003")) {
    return NextResponse.redirect("https://educationministry.org");
  }

  // Protect dashboard routes
  const token = request.cookies.get("next-auth.session-token");
  if (request.nextUrl.pathname.startsWith("/dashboard") && !token) {
    return NextResponse.redirect(new URL("/auth/login", request.url));
  }

  return NextResponse.next();
}

export const config = {
  matcher: ["/((?!api|_next/static|_next/image|favicon.ico).*)"],
};
```

11. Appendix — Quick Reference

11.1 Environment Variables

VARIABLE	DESCRIPTION	EXAMPLE
`DATABASE_URL`	PostgreSQL connection string	`postgresql://user:pass@localhost:5432/eduministry`
`NEXTAUTH_URL`	Application base URL	`https://educationministry.org`
`NEXTAUTH_SECRET`	JWT signing secret (shared across BTPMA)	`super-secret-key-shared`
`GOOGLE_CLIENT_ID`	Google OAuth client ID	`xxx.apps.googleusercontent.com`
`GOOGLE_CLIENT_SECRET`	Google OAuth secret	`GOCSPX-xxx`
`RESEND_API_KEY`	Resend email API key	`re_xxx`
`S3_BUCKET`	Resource storage bucket	`eduministry-assets`
`S3_REGION`	AWS region	`us-east-1`
`NEXT_PUBLIC_SITE_URL`	Public site URL (client-side)	`https://educationministry.org`

11.2 NPM Scripts

SCRIPT	COMMAND	DESCRIPTION
`dev`	`next dev -p 3003`	Start development server
`build`	`contentlayer build && next build`	Build for production
`start`	`next start -p 3003`	Start production server
`lint`	`next lint`	Run ESLint
`db:push`	`prisma db push`	Push schema to database
`db:migrate`	`prisma migrate dev`	Create & run migration

`db:seed`	`tsx prisma/seed.ts`	Seed database with sample data
`db:studio`	`prisma studio`	Open Prisma visual editor
`content:build`	`contentlayer build`	Build MDX content layer
`sitemap`	`tsx scripts/generate-sitemap.ts`	Regenerate sitemap

11.3 Database Models (Prisma Summary)

MODEL	KEY FIELDS	RELATIONS
`User`	id, email, name, role, avatar	→ Enrollments, Bookmarks, EventRegistrations
`Course`	id, title, slug, category, level, published	→ Lessons, Enrollments
`Lesson`	id, title, order, courseId, contentPath	→ Course, Progress
`Enrollment`	id, userId, courseId, status, enrolledAt	→ User, Course, Progress
`Progress`	id, enrollmentId, lessonId, completed	→ Enrollment, Lesson
`Event`	id, title, date, location, capacity	→ EventRegistrations
`EventRegistration`	id, userId, eventId, status	→ User, Event
`Resource`	id, title, type, category, fileUrl	→ Bookmarks
`Newsletter`	id, email, subscribedAt, active	Standalone
`YCAPApplication`	id, name, email, essay, status	→ User (optional)